

Stat 220 — Interview Gauntlet: Scenario Cards

Cut along the boxes. Two minutes per answer, out loud, no notes. Score one point per beat: clarify, assumptions, approach, traps, what would change your mind.

CARD 1

“Our A/B test shows the new button converts 40% better, $p = 0.04$. We tested 500 users. Ship it?”

Scorer’s hint: sample size, absolute vs relative, winner’s curse, was this the only test run

CARD 2

“A supplier says their part fails only 0.1% of the time, and we use three of them independently, so total failure risk is negligible.”

Scorer’s hint: independence, shared causes, at-least-one arithmetic

CARD 3

“Our fraud model flags 2% of transactions and only 1 in 20 flags is real fraud. The model is useless.”

Scorer’s hint: base rate, lift over base rate, cost of each error, threshold

CARD 4

“Sales reps who attended the training sold 18% more this quarter. Make it mandatory.”

Scorer’s hint: selection, regression to the mean, no control group, staged rollout

CARD 5

“Our average customer spends \$80 a month, so a 10% price increase adds \$8 per customer.”

Scorer’s hint: skew, mean vs median, elasticity, flaw of averages

CARD 6

“We surveyed 40,000 users and 78% love the new design. Margin of error under 0.5%.”

Scorer’s hint: response rate, nonresponse bias, big data paradox, what the MOE covers

CARD 7

“The model was 95% accurate in testing and is failing badly in production.”

Scorer’s hint: leakage, distribution shift, tuning on the test set, base-rate change

CARD 8

“Region A has a 12% complication rate, Region B has 6%. Retrain Region A’s staff.”

Scorer’s hint: denominator, case mix, standardization, sample size, intervals

CARD 9

“Adding 20 new features raised our R^2 from 0.61 to 0.78. Much better model.”

Scorer’s hint: in-sample vs out-of-sample, overfitting, cross-validation, one-standard-error rule

CARD 10

“Our overall refund rate rose, but it fell in every single product category.”

Scorer’s hint: Simpson’s paradox, mix shift, disaggregate before concluding

CARD 11

“A 99%-accurate screening test came back positive for a disease that affects 1 in 1,000.”

Scorer’s hint: Bayes, base rate, natural frequencies, what decision follows

CARD 12

“Our top-performing store slumped this quarter after we changed its manager.”

Scorer’s hint: regression to the mean, winner’s curse, control group

CARD 13

“We have 12 million rows of app telemetry, so our results represent the market.”

Scorer’s hint: sampling frame, undercoverage, bias does not shrink with n

CARD 14

“Income has 30% missing values, so I filled them with the median and moved on.”

Scorer’s hint: MCAR vs MAR vs MNAR, missingness indicator, understated uncertainty

CARD 15

“The random forest says support tickets are the top driver of churn, so let’s reduce tickets.”

Scorer’s hint: importance is not effect, reverse causation, symptom vs cause

CARD 16

“Our 95% confidence interval is 2 to 8, so there’s a 95% chance the true effect is in there.”

Scorer’s hint: CI vs credible interval, what the probability is about, when they nearly coincide

CARD 17

“Conversion dropped from 4.1% to 3.8% last month. What happened?”

Scorer’s hint: sample size, tracking or definition change, traffic mix, noise before causes

CARD 18

“We tried 200 model configurations and the best got 91% on the test set.”

Scorer’s hint: selection on the test set, nested CV, maximum of noisy scores